Dharmakrishnan Kottumukulur Radhakrishnan
Calgary, AB, Canada | 4038744038 | dharmakrishna@gmail.com

Summary

- A highly focused, results-driven IT professional with around 20 years of experience, including over a decade in the big data ecosystem. Specializing in data warehouse architecture, data integration frameworks, and data analytics.
- Hands-on in architecture, design and development of data platform in both Hadoop ecosystem and cloud platform(AWS, Azure). Lead and guide the team to accomplish the delivery in agile software development environment.
- Lead the team in implementing the security and data governance compliance set within the organization leading to SOC2 and ISO certification.
- Proficient in Data modelling and make sure data is always secured both at rest and in transit leveraging the encryption strategies.
- With the proficiency in scripting language, strongly enforce automation of repetitive task. Some of which had provided cost optimization in multi fold by automating management of unused resource and have automated code generation, which helps in developing the code by templates and increase productivity.
- Advocate the use of serverless architecture and design system with optimal performance and cost. Have achieved cost reduction of 30K USD per month by evaluating the optimal use of resource.
- Adequate knowledge of Machine learning algorithms. Acquired hands-on knowledge of LLM, Agentic framework and its application using Langchain, Langgraph, prompt engineering and vector database by implementing POC for the organization.
- ISC2 Certified in cybersecurity, With substantial knowledge of platforms Snowflake, Microsoft Fabric and hold credentials in Databricks Lakehouse Fundamentals.

Key Technical Competency

- Operating system : Windows,Solaris, Unix, Linux
- Cloud : AWS, Azure, GCP
- Platform/Languages : Python, Java, Scala, C/C++, Shell scripting, PL/SQL, PostgresSql, Map-Reduce, Hadoop, spark, Flink, AWS-EMR,AWS-ECS, Docker,Kubernetes, Bigquery, Azure Data Factory, Azure Data Fabric, Databricks
- Framework/Library : FastAPI, Pandas, numpy, Pydantic
- LLM Framework : Langchain, Langraph, MCP
- Database Technologies : Postgres,DB2,Hadoop,Hive,HBase, Vector database,duckdb,snowflake, DBT, Iceberg tables
- Tools : pgadmin, Pentaho, Coverity, Airflow, Protegrity, Jenkins, packer, Looker, kafka, Airbyte, Quicksight,Circleci, Datadog,Vanta, Snyk
- SDLC : Waterfall, Agile

Employment History

Project 10: Sharepoint Reporting

Company: SDK TEK Service Ltd(Canada)

Duration: 3 Months From**: October, 2025** To**: Till date**

Location: Calgary, Alberta

Description of the Project:

SharePoint Reporting project is to collect the sharepoint site metadata file and the org level 365 group file, ingest the data into the Datawarehouse and create a report providing insight into assessing how many sites have the sensitivity label assigned correctly and provide the usage statistics for the Data Governance.

Responsibilities:

- Design and implement the Data ingestion pipeline using Azure Data Factory to bring the data from source into the bronze and silver layer (Delta Lake)
- Design the Data Model and transform the data from silver layer into Gold layer in snowflake warehouse for reporting using PowerBI.
- Exploring Microsoft Fabric for the Data pipeline and reporting framework.
- Responsible for the Devops involved in creating the pipeline and ingesting the data.

Project 9: Validere Technologies Inc

Company: Validere Technolgies Inc(Canada)

Duration: 3 Years 4 Months From**: June, 2022** To**: October, 2025**

Location: Calgary, Alberta

Description of the Project:

Data Engineering/Platform team helps in coming up with solution and implementation of ingesting the measurement data provided by automated channels(like sftp, email, IoT etc) to the Saas application. Provided platform to onboard the entities (facility, equipment etc) of the companies/client. Build and maintain the Data warehouse platform(snowflake) and Platform Engineering for the organization.

Responsibilities:

- Led the evaluation process for choosing the Data warehouse platform between snowflake and Databricks and created the design for the ingestion process to ingest data into snowflake and perform Data modelling and provide Datashare for easy data access for customers.
- Design and implement Agentic application in Snowflake to generate insights by text for the business users/clients. Also created RAG application for product team to interact with LLM with specific domain knowledge.
- Architected and developed ingestion platform that will be used by the Technical consultant team to ingest data for onboarding of entities for clients, which increased the productivity by 30%. Also providing self serve capability for the clients.
- Voluntarily took the cost optimization initiative and achieved cost reduction of 30K USD per month by evaluating the optimal use of resource for both data platform and Data Science team without compromise on the application latency or scalability.
- Lead the security initiatives within Dataplatform and support for other teams to meet the standard compliance for ISO and SOC2.
- Designed and developed framework to enable the security team to centralize the log data from multiple accounts and provide serverless solution to ingest the centralized logs to SIEM platform.
- Designed and developed templated framework leveraging duckdb to pull data from the application and create templated reports using sql, which helped in self serve provisioning of custom templated reports by product and Consulting team.
- Handled end to end Devops lifecycle within the platform to confirm to the security and logging best practices. Also addressing the vulnerabilities and infrastructure inefficiencies.
- Design and setup Quicksight visualization tool to enable the Data Analytics users to create dashboards for the application. Also enable auto registration of user/groups using serverless framework.
- Came up with design and implementation of Disaster Recovery strategy for data and application, which can be used by all of the team within Engineering with minimal effort/change.
- Involved in the management of Vendors like AWS, Coralogix, Opsguru in driving ingestion platform, Devops and security effort across the organization.

Project 8: Benevity Inc

Company: Benevity Inc(Canada)

Duration: 1 Years 4 Months From**: Nov, 2020** To**: June,2022**

Location: Calgary, Alberta

Description of the Project:

Cloud Data Platform is the data warehouse platform that hosts all the data across Benevity Saas application ingesting in real-time and providing platform for data analysis and BI reporting. The platform is also used for data insights generation.

Responsibilities:

- Design and develop real-time streaming application using Flink. Help the team of junior developers to overcome the issue faced in teal-time streaming using Flink and kafka.
- Come up with the design for data ingestion as part of the data lake solution in GCP
- Handled the conformance of the Data security and governance within the team set by organization goals.
- Design and implement the Data Model for transformation and automated process for data validation between the source and data warehouse db(postgres) using Airflow.
- As a team lead took care of the vulnerabilities reported by Snyk on the cloud resources provisioned in CDP.
- Guide the team when team member encounters issues in streaming application, Cloud infrastructure.
- Leverage tools like openmetadata to come up with Data catalog and governance solution.

Project 7: Equifax Inc

Company: Equifax Inc (USA)

Duration: 1 Years From**: Jan, 2019** To**: Nov, 2020**

Location: Alpharetta, Georgia

Description of the Project:

Partners & Integration team is responsible for creating the platform that will be used by the Data Science and statisticians. The team performs the duty of the Data Engineering to provide the Data analysts with appropriate data in efficient way and provide the Data Scientists/Analyst with appropriate tools to perform their task efficiently in Cloud (Amazon AWS)

Responsibilities:

- Architect and lead the team in building the platform for custom Data Analytics tool that is integrated in AWS EMR cluster and helped in interfacing the tool with partner solution to enable ML feature to be accessed for credit insight derivation and action.
- Suggested and built the Data Analytics tool to be run using dockerfile, so it can be run in multiple cloud in partner solutions.
- Involved in installing the partner decisioning platform(kubernetes application) in our AWS platform.
- Implementing EKS cluster to spin up pods in secondary CIDR range, so EKS cluster can have ample ip's to create the pods.
- Designed and developed spark application to screen the consumer profile against set of credit rules to determine if the consumer is eligible for credit.
- Came up with automated process to reduce the AWS cost by stopping the services and turning it on when required, which provided cost saving of 30%.
- Manage the Vendor and Partners by leading the directions for Vendors and collaborating with Partners in design and implementation.

Project 6: Equifax Inc

Company: Equifax Inc (USA)

Duration: 3 Years 7 Months From**: May, 2015** To**: Jan, 2019**

Location: Alpharetta, Georgia

Description of the Project:

Consumer Data & Analytics team is involved in building a Data warehouse in Hadoop platform. The objective is to ingest all the data from different sources with respect to Global Consumer Services, perform data modelling on the data store to serve data for business reporting and analytics purpose.

Responsibilities:

- Design and develop ETL job using in-house tool(loadmanager) to support legacy jobs and created new process to build the pipelines and orchestrate using Airflow.
- Create Datamodel for the new data sources and update the existing model and ETL job to update the tables based on Data model.
- Came up with Automated code generation using templates and provided productivity improvement reducing 2 day work to just couple of hours.
- Designed and guide the operations team to perform the HDFS transparent Encryption to comply with DAta protection guidelines along with migration of existing encryption to protegrity Tokenization, which enhanced the security posture of the Data platform. This was achieved by coming up with UDF, which made the use of tokenization seamless to the users.
- Independently collaborated with professional services team and came up with automated solution to support the business process of extracting custom client-specific request by updating configuration files.
- Lead the project to implement the GDPR requirement to satisfy the purge and data protection compliance within the stipulated time defined for the compliance.
- Involved in creation of partner Dashboard in spotfire to provide metrics related to each partner data which helped in deriving business metrics.
- Voluntarily took up the task of performance tuning and major upgrade of Hadoop platform to completion during the critical phase of the project when the architect of the project left the project.
- Collaborate and direct the Vendor in the implementation by providing the architecture and design for the new use-cases built.

Project 5: GE

Company: SmartSource Technologies Inc(USA)

Client: GE. (USA)

Duration: 4 Months From**: Jan, 2015** To**: May, 2015**

Location: San Ramon, California

Description of the Project:

Oil & Gas Digital Framework is the project that was started with the objective to organize all the sensor data from different systems that collect data from different parts of the oil and gas machinery in the field. All the sensor data is pulled into single environment and processed to act as single source of truth that can then be used for Analytics.

Responsibilities:

- Design and develop the ETL job using Talend to ingest data from system and smartsignal into Historian. Ingest the processed data into Hadoop and analyse using Hive. This helped in the easy ingestion of data into Datalake, which is the core platform for OGDF.
- Collaborate cross-team with product manager and the backend team to understand the requirement and ingest diversified data into the Datalake.

Project 4: Cisco Systems

Company: Infosys Limited(USA)

Client: Cisco Systems. (USA)

Duration: 5 Months From**: Mar, 2014** To**: Jan, 2015**

Location: San Jose, California

Description of the Project:

Cisco Connected Analytics for Collaboration(CAC) is a product aimed at providing usage analytics of the different collaboration channels i.e webex, IM, IPPhone, Desktop & Immersive Video that are used across the organization. CAC will also provide the travel & carbon avoidance achieved by usage of the different collaboration channels, providing Organization an insight of the usage across the units/Deparment and across the different segments and geographies.

Responsibilities:

- Architect and and lead Data team to develop the ETL job using Pentaho PDI to process the data from different collaboration channels(messenger, email, etc). Design the data model to be consumed by the webservice for rendering in front-end. This solution design provided product team to expand the solution to other channels with ease of plug-and-play compatibility.
- Co-ordinate with cross team design and development to integrate the use of Data model from the Data warehouse implemented.
- Lead and co-ordinate the offshore team in the implementation and project management for the delivery of project and the platform.
- Collaborate with multiple Vendors in Cisco to drive the project and in collection of data.

Project 3: Integrated Vehicle Health Monitoring - Infosys

Company: Infosys Limited (India)

Duration: 1.2 years From**: Dec, 2012** To**: Feb, 2014**

Location: Infosys Limited**,** Chennai, India

Description of the Project:

Integrated Vehicle Health Monitoring(IVHM) is a tool built leveraging the BigData ecosystem to analyse the sensor data that is generated by aircraft system during the course of its flying time. The tool analyses the data generated by the aircraft system to diagnose the parts where the sensor values had read out of the normal range applicable for the part. The tool process the timeseries sensor data to detect out of range parts and would suggest possible cause of the failure of the part based on the sensor values. The tool enables to compare the analytic view of the sensor values between two similar part in an aircraft. The tool is intended for the aircraft manufacturing/serviceprovider to analyse all the aircraft(either similar or different make) in their fleet.

Responsibilities:

- Setup Hadoop ecosystem(Hadoop, HBase) to analyze the time series data from IoT device on flight data simulation.

- Design and develop map-reduce program to process and detect the error records in the data and project the anomaly in the time series data. This helped in showcasing the anomaly detection capability as new product service to the clients and gained traction for new client project.

Project 2: Equifax – Small Business Exchange

Company: Infosys (USA)

Client: Equifax Inc

Duration: 5.5 years From**: Jun, 2007** To**: Nov, 2012**

Location: Alpharetta, GA(USA), Chennai(India)

Description of the Project:

SBE deals about commercial solution of Credit Reporting. SBE provides the application for both consumer and commercial solutions. With extensive commercial credit data available from Equifax, it's easy to weigh the risks versus the rewards. Equifax helps to leverage the most relevant consumer information to gain insight into the customer base. Understanding a consumer's credit capacity, character and collateral issues are key to making sound risk decisions in today's economy. For many of the world's leading businesses in the financial services, retail, auto, mortgage, communications/utilities and other sectors, Equifax is part of their essential decision-making fabric and an intrinsic part of their business.

Responsibilities:

- Design and develop critical Data model migration project for Small Busines Exchange. This involved co-ordination with Architect team to design the new Data model and create framework for credit report generation.The design helped in the development of new reports from 1 month to 1 week.
- Designed the Master Data Management in Greenplum, which helped business team to come up with analytics reports and drive key business decisions.
- Involved in co-ordination of the project with offshore and help SBE platform in creating multiple credit reports and automation of the batch process to deliver business report data to clients.

Project 1: ZebOS

Company: Infosys (India)

Client: IPInfusion (USA)

Duration: 1.8 years From**: Oct, 2005** To**: May, 2007**

Location: Infosys Limited**,** Chennai, India

Description of the Project:

ZebOS is an advanced network protocol stack solution that help equipment providers bring IP-based and Ethernet-based Next Generation Network equipment to market on time, with ensured quality and reduced cost and risk. ZebOS has been deployed in the most mission-critical carrier and cloud networks. ZebOS provides a scalable, hardware-independent architecture that enables NEMs and ODMs to efficiently incorporate networking capabilities into their communications products.

Responsibilities:

- Design and development of L2 protocols(i.e RSTP, CFM etc) stack.
- Leading 3 member in the development of other L2 Protocol stack.
- Porting of the software across different OS(linux, vxworks etc).

Qualification

- Bachelor of Technology (B Tech) from Karunya Institute of Technology, Coimbatore, Tamilnadu,India